

HealthConnect Clinic

Week 5 Project Summary — Data Analytics Track

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This summary is deliberately concise and complements — rather than repeats — the full Week 5 Analytics Report (docs/HealthConnect_Week5_Analytics_Report.docx) and the executed notebook (notebooks/HealthConnect_Week5_Analytics.ipynb).

1. What We Planned to Accomplish

Per the Week 4 proposed focus: calculate the 5 KPIs identified (not just define them), deepen the exploratory analysis into variables not yet fully explored (waiting time, distance, cancellations specifically), produce a visual report, and generate business insights.

2. What Was Actually Completed

- Re-confirmed Week 4's data-quality findings still hold (no cleaning needed).
- Deepened EDA: waiting time and distance vs. outcome, cancellations analysed separately from no-shows, and a lead-time × prior-no-show interaction check.
- Calculated all 5 Week 4 KPIs with full definitions, rationale, values, and interpretation, each linked to its business question.
- Built a 4-panel analytical dashboard covering outcome distribution, lead-time effect, prior-history effect, and reminder-channel effect.
- Produced 6 business insights with practical implications for HealthConnect.
- Documented a cross-track collaboration point with the Data Science track.
- Updated the Week 4 assumptions/limitations/risks register with current status.

3. Key Findings / Development Outcomes

- No-Show Rate: 48.5%. Attendance Rate: 46.3%. Cancellation Rate: 5.3%.
- Reminder Effectiveness: 47.4% (sent) vs 51.4% (not sent) — a ~4-point reduction; SMS is the best-performing channel.
- Repeat No-Show Rate: 10.6% of appointments come from patients with 2+ prior no-shows, and that segment's no-show rate is 61.0% — well above baseline.
- Cancellations behave differently from no-shows: their rate stays flat (~5%) regardless of lead time, while no-show rate climbs steeply — supporting the Week 4 assumption that they should be treated as distinct events.
- Waiting time and distance to clinic remain weak standalone predictors, consistent with the Week 4 flag.

4. Major Challenges Encountered

- Interpreting KPI 4 (Reminder Effectiveness) required care: the raw point difference is small (~4 points), so the summary is written to avoid overstating reminders as a fix, while still recognising the value of channel choice (SMS).
- Deciding how much new EDA to add without duplicating Week 4 — resolved by scoping Week 5's EDA specifically to variables Week 4 flagged as "weak so far" or unexplored (waiting time, distance, cancellations, interactions), rather than re-running the whole Week 4 analysis.

5. Important Decisions Made and Why

- Kept Cancelled and No-Show as separate KPIs/analyses throughout, rather than combining them into one "non-attendance" metric — Week 5's evidence (flat cancellation rate vs. climbing no-show rate) confirmed this was the right call.
- Did not produce a separate cleaned/processed dataset file, since the Week 5 data-preparation check found nothing that needed cleaning — consistent with the Week 4 verdict and the brief's instruction to only save derived files when transformation actually occurs.

6. Changes to the Week 4 Approach

None. The Week 5 work followed the Week 4 proposed approach as planned — no course correction was needed.

7. Cross-Track Collaboration Completed

Shared the finalised variable-signal findings (strong: booking_lead_days, previous_no_shows; moderate: reminder_sent/reminder_channel; weak: distance_to_clinic_km, waiting_time_minutes) as a candidate-feature shortlist relevant to the Data Science track's no-show prediction model. Full detail in the Analytics Report, Section 7.

8. Remaining Work

- Statistical significance testing (chi-square, logistic regression) on the KPI drivers — descriptive only so far.
- Broader interaction analysis beyond the single lead-time $\times$ prior-no-show check done this week.
- Resolve the Week 4 open items that remain outstanding: the Sunday-appointment operating-hours question, and a documented missing-data handling rule for distance/waiting time.

9. Proposed Focus for Week 6

- Run significance tests on the confirmed KPI drivers to move from associative to statistically supported findings.
- Test additional variable interactions (e.g. reminder channel $\times$ prior history).
- Translate the 6 business insights into a stakeholder-ready recommendation with estimated impact, ready for coordination with the Data Science and Project Management tracks.

Full documentation: [docs/HealthConnect_Week5_Analytics_Report.docx](#) | Code: [notebooks/HealthConnect_Week5_Analytics.ipynb](#) | Navigation: [README.md](#)